

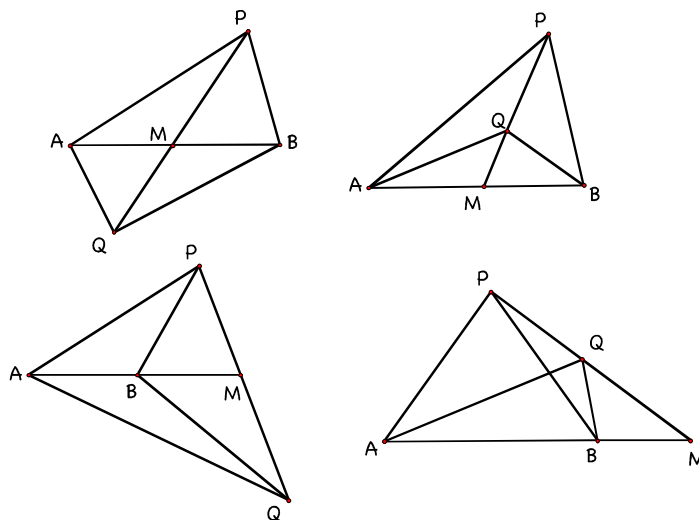
Olympiad Geometry – II

2. An useful tool -- Area

Area Lemma

If two lines PQ and AB intersect at M , then

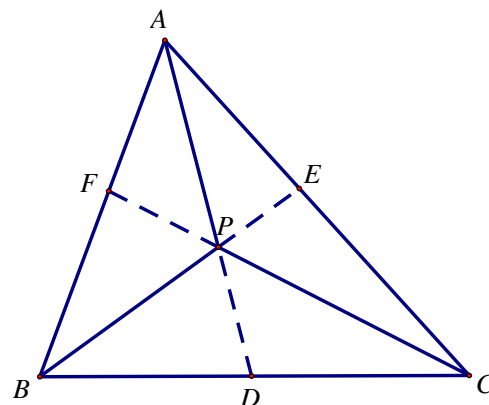
$$\frac{[PAB]}{[QAB]} = \frac{PM}{QM}.$$



Example 2.1 (van Aubel's Theorem)

In triangle ABC let D, E, F be points on the sides BC, CA, AB , respectively such that AD, BE and CF are concurrent at P . Then

$$\frac{AP}{PD} = \frac{AE}{EC} + \frac{AF}{FB}.$$



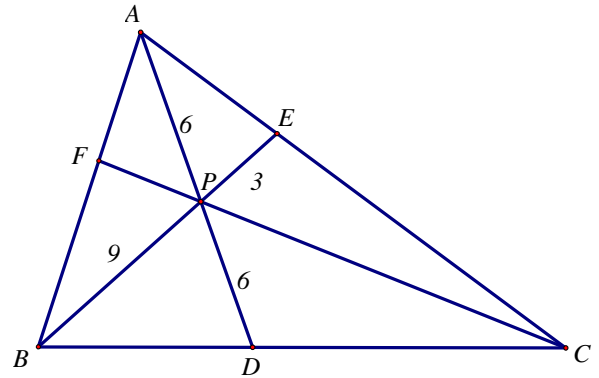
Corollary

Let ABC be a triangle with incentre I and let $AI \cap BC = D$.

$$\text{Then } \frac{AI}{ID} = \frac{b+c}{a} \text{ and } \frac{ID}{AD} = \frac{a}{a+b+c}$$

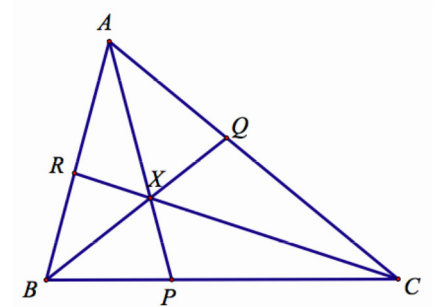
Example 2.2 (AIME 1989)

Let ABC be a triangle and let D, E, F lie on the sides BC, CA, AB respectively such that AD, BE and CF are concurrent at P . Given that $AP = 6, BP = 9, PD = 6, PE = 3$ and $CF = 20$, find the area of triangle ABC .

**3. Ceva and Menelaus****Ceva's Theorem (Concurrence of cevians)**

Let ABC be a triangle, and let P, Q, R be points on the sides BC, CA, AB respectively. Then the following assertions are equivalent:

- (a) Lines AP, BQ, CR are concurrent.
- (b) $\frac{BP}{PC} \cdot \frac{CQ}{QA} \cdot \frac{AR}{RB} = 1$
- (c) $\frac{\sin \angle PAC}{\sin \angle BAP} \cdot \frac{\sin \angle QBA}{\sin \angle CBQ} \cdot \frac{\sin \angle RCB}{\sin \angle ACR} = 1$ (trigonometric form)



Proof:

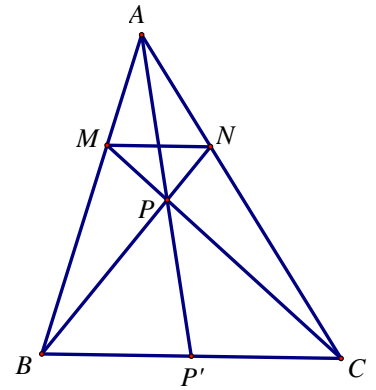
Concurrence of famous cevians

In triangle ABC the following cevians (always denote their intersections with BC, CA, AB by P, Q, R respectively) are concurrent:

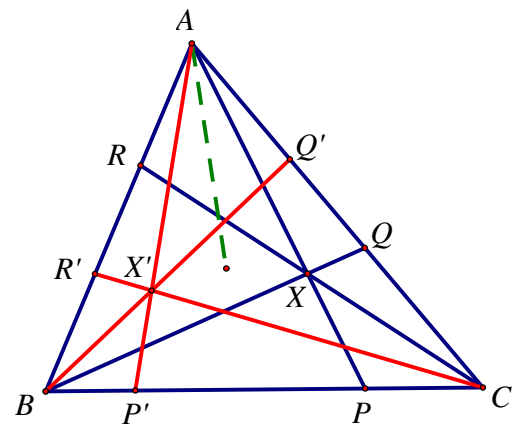
- (a) medians,
- (b) angle bisectors,
- (c) altitudes,
- (d) cevians corresponding to the points of tangency of the incircle (Gergonne point),
- (e) cevians corresponding to the points of tangency of the excircles with the triangle sides (Nagel point).

Example 3.1

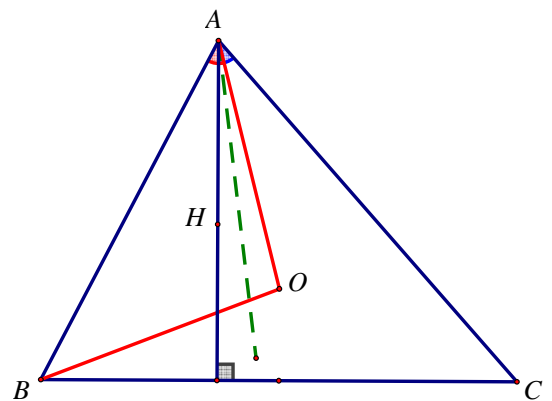
Points M, N on the sides AB, AC of the triangle ABC satisfy $MN \parallel BC$. Prove that lines BN and CM intersect on the A -median of triangle ABC .

**Example 3.2 (Existence of isogonal conjugate)**

Let cevians AP, BQ, CR concur at point X . Now construct cevians AP', BQ', CR' which are isogonal to AP, BQ, CR respectively in the respective angles. Prove that the cevians AP', BQ', CR' are concurrent. The point of concurrence is called the isogonal conjugate of X .

**Example 3.3 H and O are friends [isogonal]**

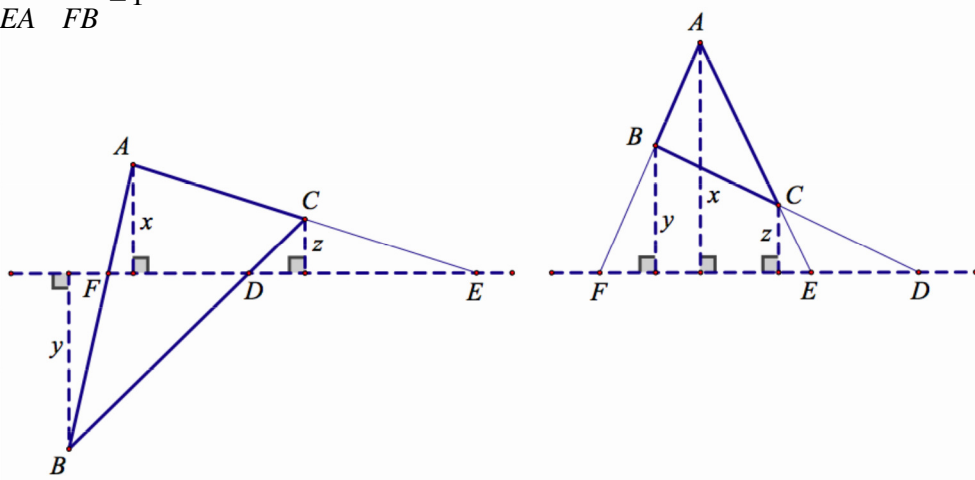
Let ABC be a triangle with orthocentre H and circumcentre O . Then the lines AO and AH are isogonal in $\angle A$ and similar result holds for pairs of lines BH, BO and CH, CO in respective angles. Therefore H and O are isogonal conjugates.



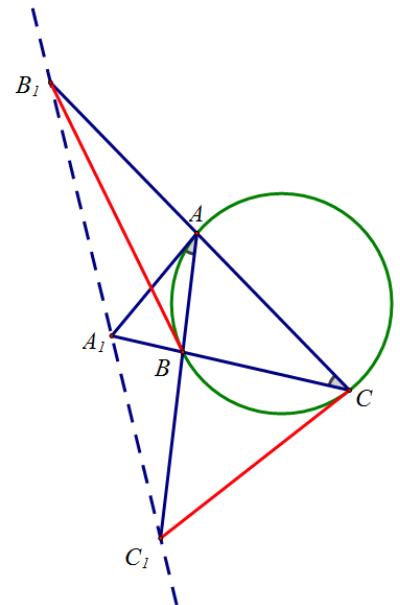
Menelaus' Theorem

Let ABC be a triangle and let points D, E, F lie on the lines BC, CA, AB respectively, so that either none or two of them lie on the triangle sides. Then the points D, E, F are collinear if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$$

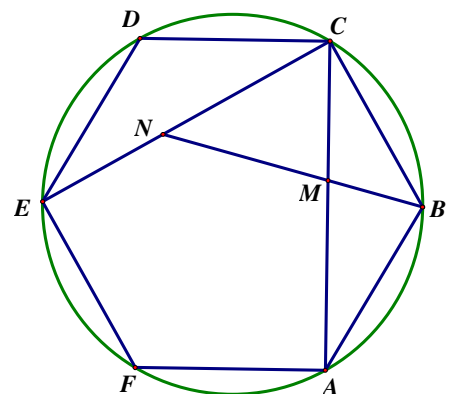
**Example 3.4**

Let ω be the circumcircle of triangle ABC and let the tangent to ω at A intersect BC at A_1 . Define points B_1, C_1 analogously. Prove that A_1, B_1, C_1 are collinear.

**Example 3.5 (IMO 1982 P5)**

The diagonals AC and CE of the regular hexagon $ABCDEF$ are divided by the inner points M and N , respectively, so

that $\frac{AM}{AC} = \frac{CN}{CE} = r$. Determine r if B, M and N are collinear.



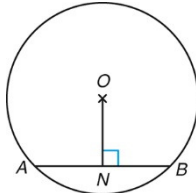
4. Simple Geometric Theorem of Circles

Chords of a Circle

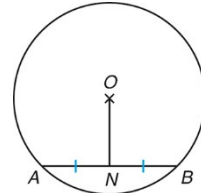
A. Perpendiculars to Chords

Theorem 4.1

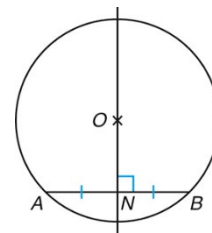
If $ON \perp AB$, then $AN = BN$.
(line from centre $\perp$ chord bisects chord)


Theorem 4.2 (Converse of Theorem 4.1)

If $AN = BN$, then $ON \perp AB$.
(line joining centre to mid-pt. of chord $\perp$ chord)


Theorem 4.3

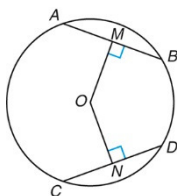
The perpendicular bisector of a chord passes through the centre of the circle.
($\perp$ bisector of chord passes through centre)



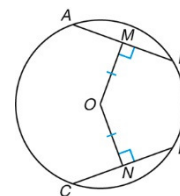
B. Distances between Chords and Centre

Theorem 4.4

If $AB = CD$, $OM \perp AB$ and $ON \perp CD$,
then $OM = ON$.
(equal chords, equidistant from centre)


Theorem 4.5 (Converse of Theorem 4.4)

If $OM = ON$, $OM \perp AB$ and $ON \perp CD$,
then $AB = CD$.
(chords equidistant from centre are equal)



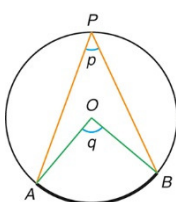
Angles of a Circle

A. Angles at the Centre and Angles at the Circumference

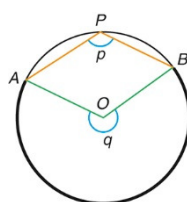
Theorem 4.6

p and q are angles at circumference and angle at centre subtended by the same arc.
Then $q = 2p$ ($\angle$ at centre twice $\angle$ at $\odot^{\text{ce}}$)

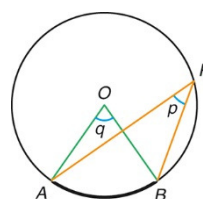
Case (i)



Case (ii)

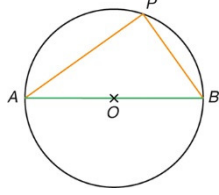


Case (iii)

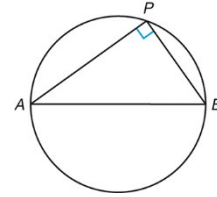


Theorem 4.7

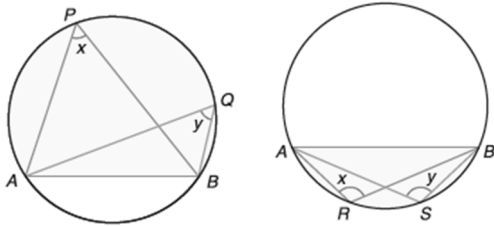
If AB is a diameter, then $\angle APB = 90^\circ$.
($\angle$ in semi-circle)

**Theorem 4.8**

If $\angle APB = 90^\circ$, then AB is a diameter.
(converse of $\angle$ in semi-circle)

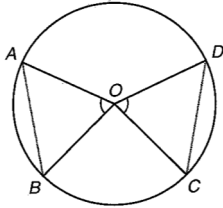
**B. Angles in the Same Segment****Theorem 4.9**

$x = y$ ($\angle$ s in the same segment)

**Relationships among Arcs, Chords and Angles****A. Equal Arcs, Equal Chords and Equal Angles****Theorem 4.10 and Theorem 4.12**

If $\angle AOB = \angle COD$, then

- (i) $\widehat{AB} = \widehat{CD}$ (equal $\angle$ s, equal arcs)
- (ii) $AB = CD$ (equal $\angle$ s, equal chords)

**Theorem 4.11 and Theorem 4.13**

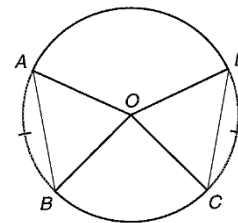
(Converse of Theorem 4.10 and 4.12)

If $\widehat{AB} = \widehat{CD}$, then

$\angle AOB = \angle COD$ (equal arcs, equal $\angle$ s)

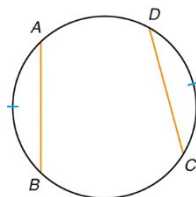
If $AB = CD$, then

$\angle AOB = \angle COD$ (equal chords, equal $\angle$ s)

**Theorem 4.14**

If $\widehat{AB} = \widehat{CD}$, then

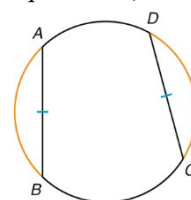
$AB = CD$ (equal arcs, equal chords)

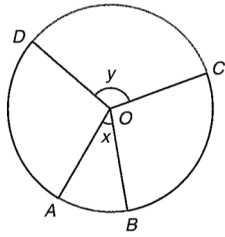
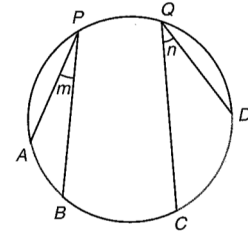
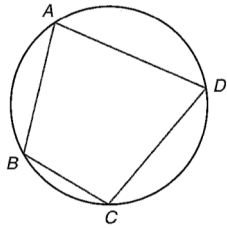
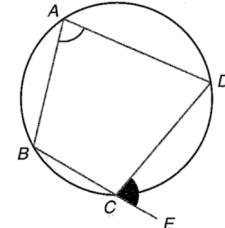
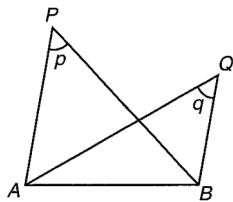
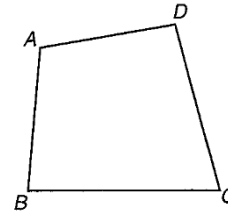
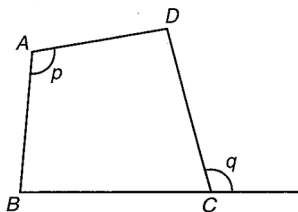
**Theorem 4.15**

(Converse of Theorem 1.14)

If $AB = CD$, then

$\widehat{AB} = \widehat{CD}$ (equal chords, equal arcs)



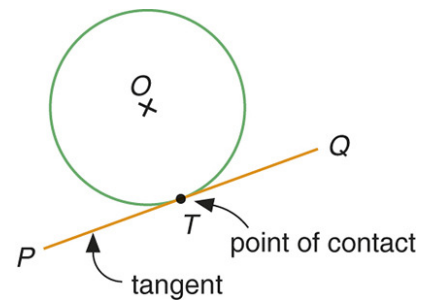
B. Arcs Proportional to Angles at the Centre**Theorem 4.16**
 $\widehat{AB} : \widehat{CD} = x : y$ (arcs prop. to $\angle$ s at centre)
**Theorem 4.17**
 $\widehat{AB} : \widehat{CD} = m : n$ (arcs prop. to $\angle$ s at $\odot^{ce}$)
**Cyclic Quadrilaterals****Theorem 4.18**
 $\angle A + \angle C = 180^\circ$ and $\angle B + \angle D = 180^\circ$
(opp. $\angle$ s, cyclic quad.)
**Theorem 4.19**
 $\angle DCE = \angle A$
(ext. $\angle$, cyclic quad.)
**Concyclic Points****Theorem 4.20**
Any three non-collinear points A , B , and C are concyclic, and there is one and only one circle, called the circumcircle of $\triangle ABC$, that can be drawn passing through them.
Theorem 4.21 (Converse of Theorem 4.9)
If $p = q$, then A , B , Q and P are concyclic.
(converse of $\angle$ s in the same segment)
**Theorem 4.22 (Converse of Theorem 4.18)**
If $\angle A + \angle C = 180^\circ$ or $\angle B + \angle D = 180^\circ$, then A , B , C and D are concyclic.
(opp. $\angle$ s supp.)
**Theorem 4.23 (Converse of Theorem 4.19)**
If $q = p$, then A , B , C and D are concyclic.
(ext. $\angle$ = int. opp. $\angle$)


Tangents to Circle and their Properties

Definition

In the figure, PQ intersects the circle at only one point T . We say that PQ touches the circle at T .

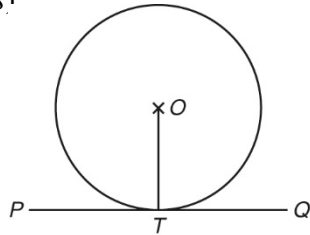
PQ is called the tangent to the circle at T , and T is called the point of contact.



Theorem 4.24

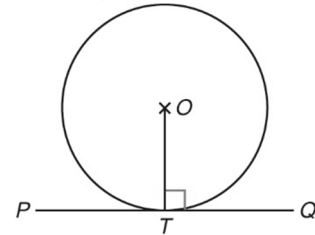
If PQ is the tangent to the circle at T , then $PQ \perp OT$.

(*tangent $\perp$ radius*)



Theorem 4.25 (Converse of Theorem 4.24)

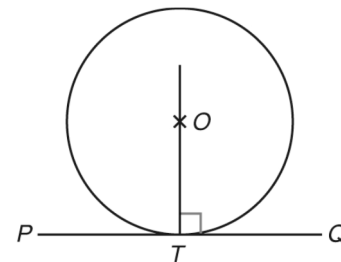
If $PQ \perp OT$, then PQ is the tangent to the circle at T . (*converse of tangent $\perp$ radius*)



Theorem 4.26

The perpendicular to a tangent PQ at its point of contact T passes through the centre O of the circle.

(*$\perp$ to tangent at its point of contact passes through centre*)



Tangents from an External Point

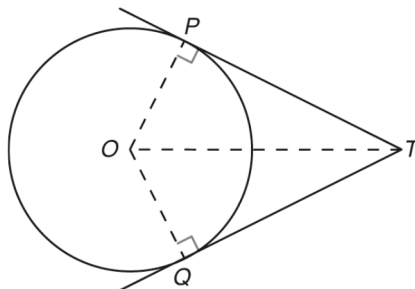
Theorem 4.27

If two tangents, TP and TQ , are drawn to a circle from an external

point T and touch the circle at P and Q respectively, then

- (i) $TP = TQ$,
- (ii) $\angle POT = \angle QOT$,
- (iii) $\angle PTO = \angle QTO$.

(*tangent properties*)



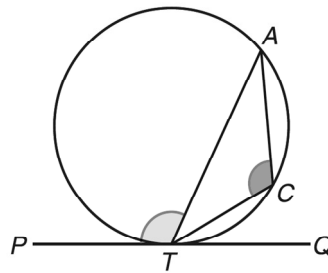
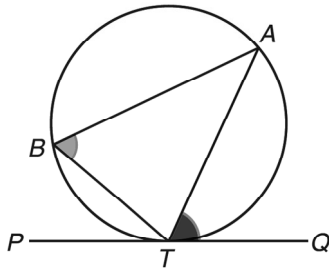
Angles in the Alternate Segment

Theorem 4.28

A tangent-chord angle of a circle is equal to an angle in the alternate segment.

(i) $\angle ATQ = \angle ABT$

(ii) $\angle ATP = \angle ACT$



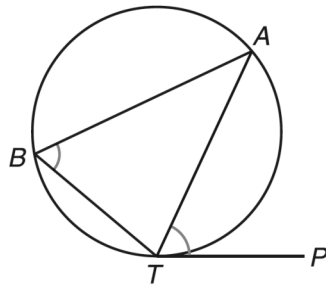
($\angle$ in alt. segment)

Theorem 4.29

(Converse of Theorem 4.28)

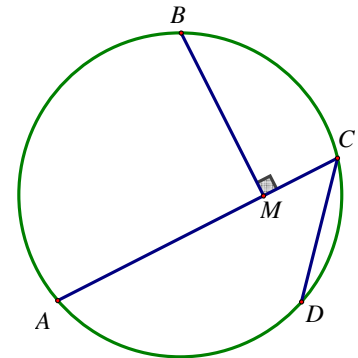
If TP is a straight line such that $\angle ATP = \angle ABT$, then TP is the tangent to the circle at T .

(converse of $\angle$ in alt. segment)



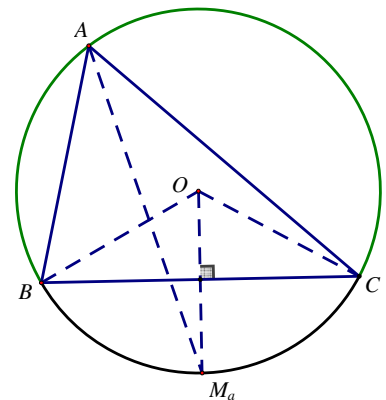
Example 4.1 (Archimedes' Split Chord)

Points A, B, C and D lie, in that order, on the circumference of a circle such that $\text{arc } AB = \text{arc } BD$. Let M be the foot of the perpendicular from B onto AC . Prove that $AM = DC + CM$.



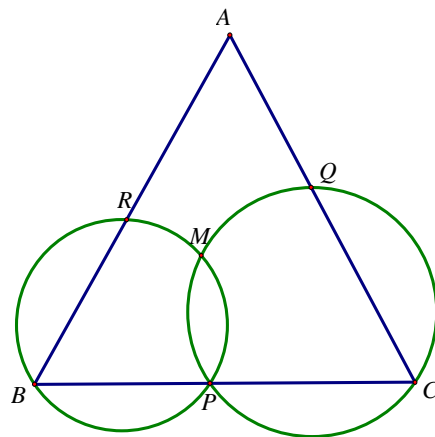
Example 4.2

Let ABC be a triangle inscribed in a circle ω and let M_a be the midpoint of arc BC of ω that does not contain point A . Then the internal angle bisector of $\angle A$ and the perpendicular bisector of BC pass through M_a .

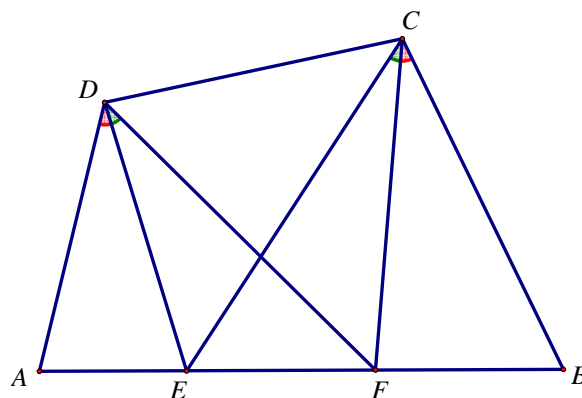


Example 4.3 (Miquel's pivot theorem)

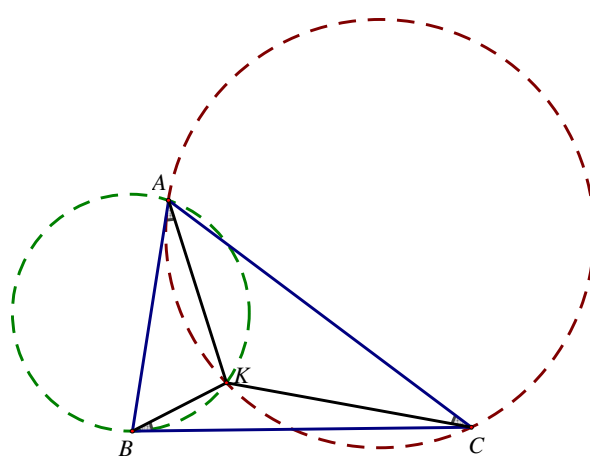
Let P, Q, R be arbitrary points on the sides BC, CA, AB of a triangle ABC . Show that the circumcircles of triangles ARQ, BPR and CQP pass through a common point.

**Example 4.4 (All-Russian Olympiad 1996)**

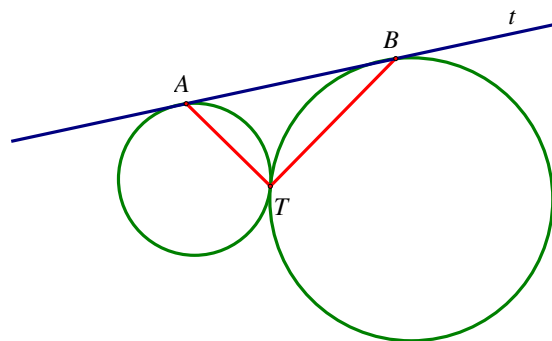
Points E and F are chosen on the side AB of a convex quadrilateral $ABCD$ such that $AE < AF$. Given that $\angle ADE = \angle FCB$ and $\angle EDF = \angle ECF$, prove that $\angle FDB = \angle ACE$.

**Example 4.5 [Angle in alternate segment]**

Let ABC be a triangle. Denote by Γ_a the circle tangent to AB at A and passing through C . Similarly, denote by Γ_b the circle tangent to BC at B and passing through A , and Γ_c the circle tangent to CA at C and passing through B . Prove that the circles $\Gamma_a, \Gamma_b, \Gamma_c$ intersect at one point.

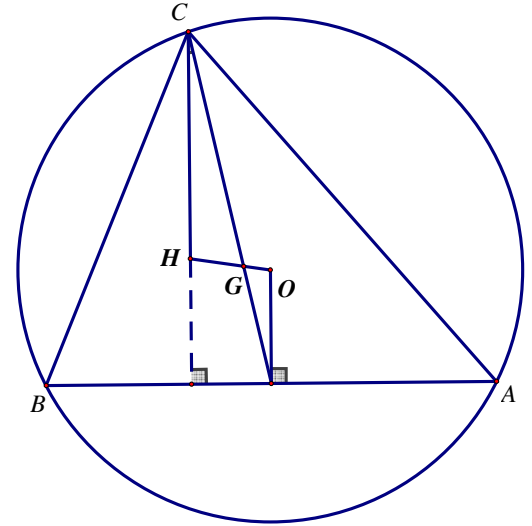
**Example 4.6 [Tangents]**

Circles ω_1 and ω_2 are tangent externally at point T . Their common external tangent t is tangent to them at A, B respectively. Show that $\angle ATB = 90^\circ$.



Euler Line

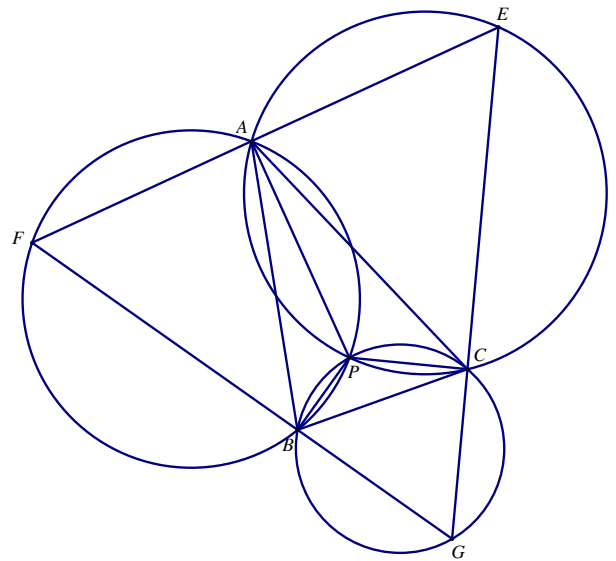
Let ABC be a non-equilateral triangle and let H, G, O be its orthocentre, centroid and circumcentre respectively. Then the points H, G, O lie on a single line (called the Euler line of the triangle ABC) in this order, and



Exercise

1. (2019 PCIMC Final Event (Senior) #20)

Let P be a point inside $\triangle ABC$ such that $\angle APB = \angle BPC = 120^\circ$ and $AP = 8$, $BP = 3$, $CP = 4$. Point D, E and F are chosen on the circumcircles of $\triangle BPC$, $\triangle CPA$ and $\triangle APB$ respectively such that DCE, EAF and FBD are straight lines. Find the greatest possible area of $\triangle DEF$.



2. (The 23rd Hong Hong (China) Mathematical Olympiad)

Let $ABCD$ be a cyclic quadrilateral inscribed in a circle Γ such that $AB = AD$. Let E be a point on the segment CD such that $BC = DE$. The line AE intersects Γ again at F . The chords AC and BF meet at M . Let P be the symmetric point of C about M . Prove that PE and BF are parallel.

3. **(IMO 2018 P1)**

Let Γ be the circumcircle of acute-angled triangle ABC . Points D and E lie on segments AB and AC , respectively, such that $AD = AE$. The perpendicular bisectors of BD and CE intersect the minor arcs AB and AC of Γ at points F and G , respectively. Prove that the lines DE and FG are parallel (or are the same line).

4. **(IMO 2020 P1)**

Consider the convex quadrilateral $ABCD$. The point P is in the interior of $ABCD$. The following ratio equalities hold:

$$\angle PAD : \angle PBA : \angle DPA = 1 : 2 : 3 = \angle CBP : \angle BAP : \angle BPC.$$

Prove that the following three lines meet in a point: the internal bisectors of angles $\angle ADP$ and $\angle PCB$ and the perpendicular bisector of segment AB .

